

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

May 2012

## IT207 Computer Networking-I

Date: 16.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

### Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

### SECTION - I

- Q - 1 (a) Explain the different criteria for any network to be considered as effective and efficient network. [04]
- (b) Define Protocol. Explain the key elements of protocol. [03]
- Q - 2 (a) Explain the Line Configuration in network. [04]
- (b) Suppose you add two new devices to an existing five-device network. If you have a fully connected mesh topology, how many new cable lines are needed? If, however, the devices are arranged in a ring, how many new cable lines are needed? [04]
- (c) Explain the different methods of Flow control. [06]

### OR

- Q - 2 (a) List and Explain the different transmission mode in network with example. [04]
- (b) Assume seven devices are arranged in a mesh topology. How many cables are needed? How many ports are needed for each device? [04]
- (c) Explain Sliding Window ARQ with example. [06]
- Q - 3 (a) Draw the architecture of OSI Layer. Explain each in detail. [10]
- Q - 3 (b) What is the difference between a service-point address, a logical address and a physical address? [03]
- (c) List the different categories of network and give one example of each category. [01]

### OR

- Q - 3 (b) What is the difference between network layer delivery and transport layer delivery? [03]
- (c) Explain the categories of data communication standards in brief. [01]

### SECTION - II

- Q - 4 (a) Explain how CRC is used in detecting errors for the following polynomial,  $g(x) = x^4 + x + 1$ . Consider the information sequence 1101011011. Find the CRC corresponding to this sequence. [04]
- (b) Generate Single bit error correcting code for following data 11011111. [03]

- Q - 5 (a) Define multiplexing. Discuss the need for multiplexing in networked system. [03]  
 (b) Explain Analog to Digital conversion in Detail. [06]  
 (c) Explain Twisted-pair cable in detail. [05]

OR

- Q - 5 (a) Explain ADSL in detail. [03]  
 (b) Explain 8-QAM with constellation diagram. Compute the bit rate for 1000 baud 8-QAM signal. [06]  
 (c) Explain Optical Fiber cable in detail. [05]

- Q - 6 (a) Define the following term: [06]  
 1. Aperiodic Signal  
 2. Carrier Signal  
 3. Baud rate  
 4. Refraction  
 5. Line discipline  
 6. ARQ  
 (b) Given Bandwidth range of 2000Hz to 12000 Hz. Draw the full duplex ASK diagram of the system. Find the carrier frequency and bandwidth in each direction. [04]  
 (c) Explain bit stuffing in TDM. What is the relationship between the number of slots in a frame and the number of input lines for synchronous and asynchronous TDM? [04]

OR

- Q - 6 (a) Define the following term: [06]  
 1. Topology  
 2. Internetwork  
 3. Transmission mode  
 4. Amplitude  
 5. Phase  
 6. Modulation  
 (b) State the Nyquist theorem. If the frequency spectrum of a signal has a bandwidth of 500 Hz with the highest frequency at 600 Hz, What should be the sampling rate according to the Nyquist theorem? [04]  
 (c) Discuss LRC and types of errors that it can detect and it cannot detect. [04]

\*\*\*\*\*